

BABA YAGA'S HUT

A SMALL CABIN. A WALKING FOOTPRINT.

Cradle-to-assembly reconstruction of a Slavic walking forest hut.



STEP ONE

ONE HUT. ONE THRESHOLD.

The model rebuilds Baba Yaga's walking forest cabin as one physical hut, delivered and commissioned once. Folklore supplies the form; engineering supplies the bill of materials.

Functional unit

One walking hut

Reference flow

1 reconstructed cabin

Time context

30-year service

Modeled event

10 km relocation

Model status

Narrative engineering reconstruction. Not a verified LCA or comparative assertion.



The functional unit is one reconstructed hut, not the entire legend.

STEP TWO

THE BOUNDARY TURNS.

INCLUDED	TREATMENT
Wood cabin materials	Modeled
Metal leg bracing	Modeled
Brick chimney	Modeled
Workshop electricity	Modeled
Delivery and installation	Modeled



EXCLUDED	RULE
Magic propulsion	Cut off
Witch operations	Cut off
Visitor use	Cut off
End-of-life	Cut off

CUTOFF STATEMENT

Cut-off: the model stops after assembly and commissioning; magic is reported as a narrative proxy, not a fuel.



The boundary includes physical assembly and excludes the supernatural driver.

STEP THREE

A CABIN WITH TALONS.



3.6 m length

PROPERTY	VALUE
Outer dimensions	3.6 x 3.0 m
Timber mass	4.80 t
Metal bracing	0.45 t
Chimney / hearth	0.55 t
Total modeled mass	5.87 t

GEOMETRY PROXY

Simplified log-cabin envelope with two structural bird-leg supports.



The model freezes geometry before applying factors or sensitivities.

STEP FOUR

THE INVENTORY WALKS FIRST.

FLOW	QTY	UNIT	EF
Wood logs, roof shingles and floor	4.8	t	S1
Metal hardware and leg braces	0.45	t	S2
Brick chimney and hearth	0.55	t	S3
Small shuttered glass window	0.035	t	S4
Bindings, seals and leather-like straps	0.03	t	S5
Workshop electricity	850	kWh	S6
Delivery freight to forest clearing	879.75	t·km	S7
Installation diesel for lifting and clearing	55	l	S8

EXPECTED HOTSPOT

The metal bracing is small in mass but high in emission factor.



Every inventory line has one factor or an explicit treatment.

STEP FIVE

BUILDING THE IMPOSSIBLE.



- 1 Fell and mill structural logs
- 2 Cut roof shingles and floor boards
- 3 Forge brackets, cuffs and claw supports
- 4 Build chimney and fit window
- 5 Assemble, brace and commission

INSIDE BOUNDARY

All stages occur before the hut enters use.

Manufacture is modeled as material procurement plus workshop electricity.

STEP SIX

ENERGY AT THE CLEARING.



Workshop electricity

850 kWh

Delivery distance

150 km

Freight activity

879.8 t·km

Installation diesel

55 l

COMMISSIONING NOTE

The walking function is modeled through materials and commissioning, not through a magical fuel.



Transport is calculated as mass times distance, not distance alone.

SOURCE LEDGER

FACTORS, PROXIES, LIMITS.

ID	FLOW	FACTOR	SOURCE
S1	Wood	269.504 kg CO ₂ e/t	DEFRA 2026
S2	Metals	3,821.949 kg CO ₂ e/t	DEFRA 2026
S6	Electricity full	0.184 kg CO ₂ e/kWh	DEFRA 2026
S7	HGV freight	0.123 kg CO ₂ e/t-km	DEFRA 2026
S8	Diesel incl. WTT	3.195 kg CO ₂ e/l	DEFRA 2026



PROXY DISCLOSURE

Leather-like straps use the average plastics factor. Magic, biological motion and visitor behavior are cut off.



DEFRA 2026 is prioritized; proxies are declared where exact factors do not exist.

BASELINE RESULT

THE LEGS DOMINATE.

3.73 t CO₂e

baseline per reconstructed hut



BASELINE CONTRIBUTIONS (RANKED)

Share of total and t CO₂e per reconstructed hut

			SHARE	t CO ₂ e
1		Metal hardware	46.1%	1.72
2		Wood logs, roof shingles and floor	34.7%	1.29
3		Workshop electricity	4.2%	0.16
4		Brick chimney and hearth	3.6%	0.13
5		Bindings, seals and leather-like straps	2.5%	0.10
6		Small shuttered glass window	1.3%	0.05

CONTRIBUTOR	SHARE	t CO ₂ e
Metal hardware	46.1%	1.72
Wood logs, roof shingles and floor	34.7%	1.29
Workshop electricity	4.2%	0.16
Brick chimney and hearth	3.6%	0.13
Bindings, seals and leather-like straps	2.5%	0.10
Small shuttered glass window	1.3%	0.05

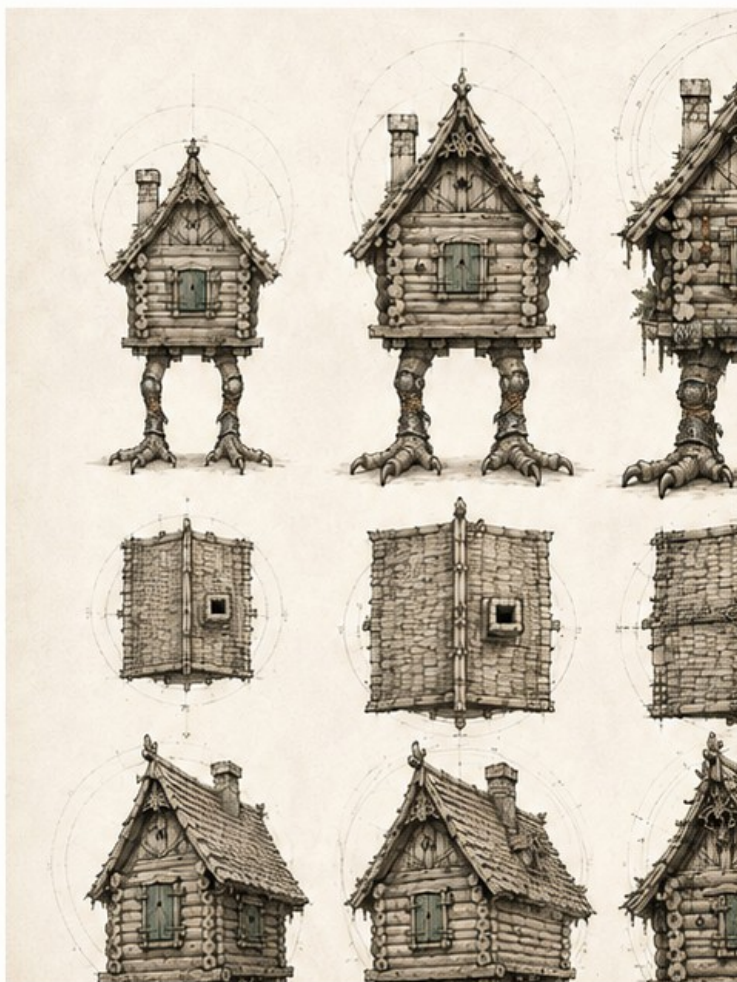
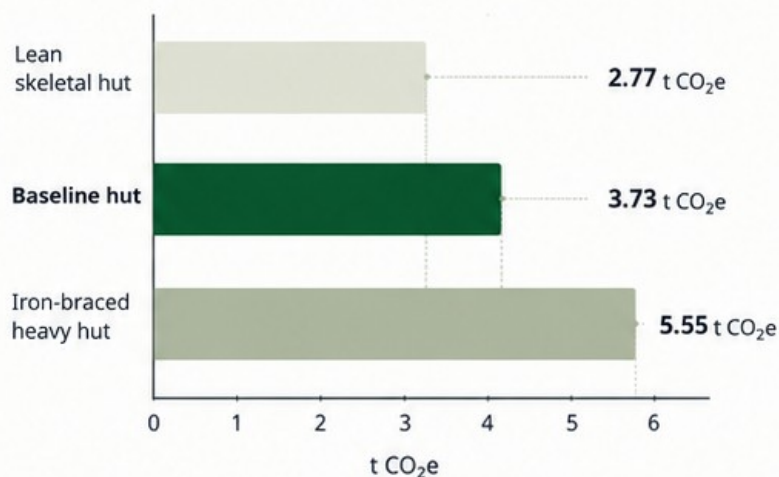
HOTSPOT

Metal bracing and hardware represent 46.1% of the baseline result.

The walking structure matters more than the wooden cabin shell.

SENSITIVITY A

MASS CHANGES THE MYTH.

EMBODIED CARBON (t CO₂e)RANGE: 2.77 – 5.55 t CO₂e

SCENARIO		CHANGE (COMPARED TO BASELINE)
	Lean	less wood, metal, energy
	Baseline	frozen page 09 value
	Heavy	more wood, metal, energy

RANGE

Physical reconstruction changes the result from 2.77 to 5.55 t CO₂e.

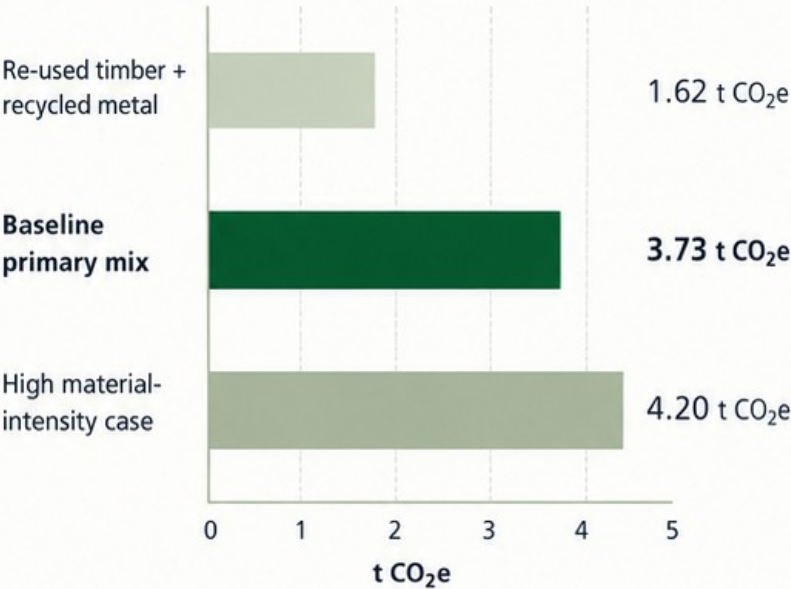
Mass is the largest modeling uncertainty under the frozen boundary.

SENSITIVITY B

SOURCE CHANGES THE STORY.



MODELED FOOTPRINT (t CO₂e)



CASE	MODEL RULE
Reuse	re-used wood + closed-loop metal
Baseline	primary material factors
High	wood +10%, metal +20%

SOURCE EFFECT

Material sourcing shifts the modeled footprint from 1.62 to 4.20 t CO₂e.



The same hut tells a different story when material origin changes.

FINDING

WHAT THE MODEL TELLS US.

01

The impossible is structural.

Metal bracing is the largest modeled hotspot because the hut must stand on moving legs.

02

Wood is visible, not dominant.

The log cabin accounts for 34.7% of the footprint under DEFRA primary wood factors.

03

Mass remains the main uncertainty.

Physical scaling changes the total from 2.77 to 5.55 t CO₂e.

04

Reuse changes the ending.

Re-used timber and recycled metal reduce the modeled total to 1.62 t CO₂e.



"A moving threshold carries a structural footprint."



A moving threshold carries a structural footprint.

SEASON II

WHERE MYTH MEETS INVENTORY



EPISODE #53 | BABA YAGA'S HUT

A small cabin can carry a structural footprint.